Compute Dilution in Budgeted Online RGB-D Gaussian Splatting

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Abstract

Online RGB-D 3D Gaussian Splatting (3DGS) spends a fixed per-frame optimization budget while its map keeps growing. We study ***compute dilution****: as the map grows, established Gaussians receive a shrinking share of the updates. Interventions on two TUM and two Replica scenes (150 frames, eight seeds each) support this as a compute-allocation effect. Raising the optimization budget in proportion to map size, with the spawn allowance fixed, improves raw-error selection on all four scenes (*$+1.56$ *to* $+3.73$ dB*); with free map growth, PSNR rises monotonically with budget and tracks the fraction of mature Gaussians updated per frame (per-seed Spearman* $+1.00$ *in every scene). The usual remedy, throttling new Gaussians, is scene dependent: coverage throttling improves raw-error selection on the TUM scenes (*$+3.53$ *and* $+1.71$ dB*) but lowers it on Replica (*-0.68 *and* -0.87 dB*). A throttled policy with temporal-influence selection reaches* 27.74 dB *on* `fr2_xyz`*,* 2.36 dB *above the strongest budgeted baseline and* 0.15 dB *below unconstrained optimization, with a* $3.2\times$ *smaller map, but it trails raw-error selection on 10 of 14 sequences, including all eight Replica scenes. Coverage also changes densification thresholds and hard caps remove capacity on Replica, so map size is not isolated as the cause. The* 15 ms *limit is modeled optimization time rather than end-to-end latency (about 10 FPS measured), and seeds measure repeatability on fixed trajectories, not generalization across scenes. Auxiliary analyses (counterfactual Gaussian utility, a learned utility model, contextual re-ranking, and a hole-rate policy router) are in the supplement.*

1. Introduction

3D Gaussian Splatting (3DGS) [6] underlies online RGB-D mapping systems such as SplaTAM [5], Point-SLAM [13], MonoGS [10], Photo-SLAM [4] and RTG-SLAM [12]. Deployed on robots and headsets, these systems have only milliseconds of optimization per streaming frame. At the same time, online pipelines add Gaussians in response to rendering error, so the map grows while the per-frame budget stays fixed; in our runs some maps exceed 100k primitives within 150 frames.

We hypothesize **compute dilution**: a growing map spreads a fixed optimization budget over more Gaussians and over newly spawned ones, so established (mature) Gaussians are updated less often and quality stagnates or degrades. We test this with interventions: a corrected-pose selector$\times$throttle factorial, optimization-budget controls (a fixed $\times 4$ budget, a budget proportional to map size, and a $\times 1/\times 2/\times 4$ dose-response) that hold the spawn allowance fixed, hard map caps, and per-frame logs of how the budget is allocated by Gaussian age, on two TUM and two Replica scenes. Raising the budget in proportion to map size improves PSNR on all four scenes ($+1.56$ to $+3.73$ dB), and with free growth the fraction of mature Gaussians updated per frame tracks quality in the dose-response. The standard remedy, throttling spawns, is not universal: it adds $+3.53$ and $+1.71$ dB on the TUM scenes but lowers PSNR on Replica. Coverage throttling also changes densification thresholds, and hard caps remove capacity on Replica, so we claim a compute-allocation mechanism under the tested protocol, not that map size is the unique cause.

Contributions. (1) A measurable formulation of compute dilution, with age-resolved allocation statistics (budget share spent on young Gaussians, selection rate of mature Gaussians) logged per frame. (2) Interventional evidence on four scenes (two TUM, two Replica; eight seeds each): a 2×3 selector–throttle factorial, fixed and map-proportional budget controls, a budget dose-response, and map-cap controls, with their limits stated. (3) A simple coverage-throttled policy that reaches 27.74 dB on `fr2_xyz` within 0.15 dB of unconstrained optimization at a $3.2\times$ smaller map, together with a scene-by-scene account of where it fails (10 of 14 sequences, all Replica scenes), including a Replica factorial showing that throttling itself lowers PSNR there. (4) Auxiliary negative and descriptive results (counterfactual single-Gaussian utility, a learned utility model, contextual re-ranking, a hole-rate router), reported in the supplement with frozen artifacts.

2. Related Work

Online 3DGS SLAM and dense mapping. GS-SLAM [19] and SplaTAM [5] introduced differentiable 3D-Gaussian SLAM, with SplaTAM using silhouette-guided optimization; Gaussian-SLAM [21] adds a sub-map design and Loop-Splat [23] adds loop closure by registering Gaussian submaps; Point-SLAM [13] uses dynamic neural point clouds; MonoGS [10] and Photo-SLAM [4] push toward real-time operation with loop closure; RTG-SLAM [12] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 5). **Scalable and efficient Gaussian mapping.** Recent systems address online map scalability through representation and update design. VTGaussian-SLAM [2] ties Gaussians to RGB-D views and avoids keeping every primitive learnable, targeting large scenes and memory use. SEGS-SLAM [18] improves structure and appearance consistency, while DyGS-SLAM [3] focuses on dynamic-scene localization and reconstruction. In parallel, SteepGS [17] derives compact density-control updates for 3DGS reconstruction. A recent online-mapping preprint studies geometry-aware initialization and densification under short optimization schedules [8]. These methods make clear that map representation, density control, and online optimization are coupled. Our study asks how growth rate and update policy should respond to coverage and observable scene statistics under a fixed per-frame budget and Gaussian representation. We do not claim density control or efficient Gaussian mapping as new in general. The per-scene SOTA numbers in Appendix Supp. Tab. S2 are protocol-mismatched context. **Density control for 3DGS.** Taming 3DGS [9] uses guided, constructive densification under resource limits; Mini-Splatting [1] targets a constrained Gaussian count; Pixel-GS [22] reweights the densification gradient by covered pixels; and 3DGS-MCMC [7] replaces heuristic densification and pruning with relocation. These methods address offline reconstruction. Our setting differs in that a per-frame optimization budget is shared between existing and newly spawned Gaussians. We do not benchmark against them, which is a limitation of the present comparison (Sec. 5). **Evaluation protocols in dense SLAM.** Published systems report full-sequence train-view rendering with global refinement at full resolution (Replica 33–42 dB for MonoGS/Point-SLAM/RTG-SLAM; TUM fr2_xyz 16–25 dB for MonoGS/SplaTAM/Photo-SLAM) plus full-trajectory ATE. Our protocol is deliberately narrower — budgeted online mapping on fixed horizons — so cross-paper numbers are contextual range, never superiority claims (Supp. Tab. S2). **Budgeted and utility-driven selection.** Error-only, error×influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers. Their ordering differs between the evaluated TUM and Replica scenes (Sec. 4). **Submodular maximization and knapsack allocation.** Budgeted set selection is related to monotone submodular maximization, for which approximation guarantees exist under explicit assumptions and appropriate algorithms [11, 16]. Those assumptions are not established for our measured reconstruction objective, and our fractional-knapsack ranker does not inherit those guarantees. The measured interactions vary by protocol, so we treat pointwise ranks as references and test map-growth control empirically.

3. Setting and Method

Budgeted mapping. Let $G_t = \{g_1, \dots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t. Per frame, a scheduler chooses a subset $A_t \subseteq G_t$ to optimize,

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \le B_t, \tag{1}$$

where ΔQ is the reconstruction improvement, approximated at runtime by a selector score, and $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ is a decoupled cost model for K microsteps. The budget B_t is *modeled* optimization time. **Measuring dilution.** We log two age-resolved statistics per frame: the *young budget share*, the fraction of modeled update budget spent on Gaussians younger than ten frames, and the *mature selection rate*, the fraction of Gaussians at least ten frames old that are selected. Dilution predicts a falling mature selection rate as the map grows. **Selection.** Candidates are scored by error × influence mass, optionally with temporal EMA filtering ($\beta = 0.90$), and packed by fractional-knapsack value density s_i/C_i under $B_t = 15\,\text{ms}$ with safety factor 1.10 (effective cap 13.6 ms). Each selected Gaussian receives $K = 5$ microsteps; the cost model is calibrated so modeled and wall-clock optimization time agree within measurement noise on the RTX 4050 backend. Exact Gaussian-to-pixel attribution costs 523.9 ms/frame, so we use a center-sampled projection, $\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4}\, e(\lfloor \mu_{2D,i} \rceil)$, that runs in 0.85 ms (614.5× faster) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. The scheduler totals 2.54 ms per frame. **Coverage throttling.** Coverage κ_t is the fraction of valid pixels with transmission < 0.5. The per-frame spawn cap ramps from 1.0 to 0.20 as κ_t goes from 0.80 to 0.90. A backlog limit exists but was inactive in the headline runs, because its count was only computed when warmup was enabled; the factorial evaluates it independently.

Warmup pre-allocation is disabled and microsteps are fixed at $K = 5$.

4. Experiments

Protocol. TUM RGB-D [15] and Replica [14] sequences at 320×240, $B_t = 15\,\text{ms}$, seeds [42..49]; two-sided paired Wilcoxon tests, Holm correction and bootstrap CIs over per-seed trajectory summaries; production `gsplat` backend; ground-truth poses with tracking decoupled (a mapping benchmark, not SLAM ATE). With eight seeds the smallest attainable two-sided Wilcoxon p is 0.0078, and with fixed trajectories seeds measure repeatability only, so we report effect sizes with intervals and do not treat seed-level standardized effects as evidence of generalization. All results in Tabs. 1–4 come from one code revision (git tag `paper-rerun-1`) with the corrected evaluator: standard Gaussian-window SSIM and alpha-normalized rendered depth. Earlier bundles used a global-statistics SSIM and a depth evaluator that biased depth L1 toward zero in partially transparent pixels, so their SSIM and depth values are not comparable with these. Artifacts: `results/rerun_paper1/`.
Holdout protocol. Because online mapping optimizes on the evaluated trajectory, train-view PSNR overstates generalization. We hold out every 8th frame render-only (never mapped, never selected): the train–holdout gap is $+0.47\,\text{dB}$ mean over 30f×3 seeds for both OURS and `error_only` (`results/phase14_holdout/`). This is a same-trajectory proxy, not independent novel-trajectory generalization.

Main result (Tab. 1, Fig. 2, Fig. 3). OURS reaches $27.74 \pm 0.03\,\text{dB}$: $+2.36$ over the strongest budgeted baseline, $+4.21$ over the RTG-SLAM-policy re-implementation, and $+8.89$ over pass-through. These results evaluate the combined policy on this trajectory.

Selector × throttle factorial (Tab. 2). With selector fixed to raw error, coverage adds $+3.53$ $[3.38, 3.70]$ dB on `fr2_xyz` and $+1.71$ $[1.64, 1.80]$ dB on `fr1_xyz`. With temporal×influence fixed, its gains are $+2.49$ $[2.40, 2.59]$ and $+1.50$ $[1.48, 1.52]$ dB. Brackets are paired seed-bootstrap 95% CIs; all four coverage contrasts have two-sided paired Wilcoxon $p = .0078$. Temporal selection itself adds $+0.09$ dB with coverage on `fr2_xyz` and $+0.54$ dB on `fr1_xyz`; the interaction is negative (-1.04 and -0.22 dB), so selector and coverage effects are not additive. The independent backlog option adds $+0.01/-0.02$ dB on `fr2_xyz` and $+0.08/+0.11$ dB on `fr1_xyz` for error/temporal selection.

Dilution controls and direct allocation. Holding the spawn allowance at its $15\,\text{ms}$ value, giving raw-error selection $60\,\text{ms}$ for optimization raises PSNR by $+3.69$ $[3.55, 3.83]$ and $+2.94$ $[2.89, 3.01]$ dB on `fr2_xyz`/`fr1_xyz`, despite maps of 71k/79k. Capping the error-only map at 23k adds $+2.76$ $[2.62, 2.90]$ and $+1.00$ $[0.96, 1.05]$ dB, but multiplies depth L1 by 2.8 on `fr1_xyz`. A 39k cap, close to that scene's coverage map size, adds $+0.04$ $[-0.03, 0.10]$ dB there (and $+0.80$ $[0.67, 0.93]$ dB on `fr2_xyz`). Thus a static size limit does not reproduce coverage throttling's effect on `fr1_xyz`. For raw-error selection on `fr2_xyz`, coverage reduces the share of modeled update budget spent on Gaussians younger than ten frames from 22.7% to 9.5%, while the per-frame selection rate among mature Gaussians rises from 37.9% to 87.4%. The corresponding mature selection rate on `fr1_xyz` rises from 28.8% to 57.4%. Allocation shifts and the budget intervention support compute dilution under the tested protocol, but coverage also changes densification thresholds and the optimization trajectory. Map size alone is not identified as the unique cause.

Exploratory transfer to two more TUM scenes (150f×3 seeds, same six factorial cells, earlier code revision): coverage adds $+0.34/+0.10\,\text{dB}$ to error/temporal selection on `fr1_desk`, but depth L1 worsens from 0.0481/0.0476 to 0.0590/0.0595 m. On `fr3_static`, coverage adds $+0.77/+0.38\,\text{dB}$ and slightly improves depth L1. Adding backlog to temporal selection on `fr1_desk` loses 0.59 dB. These scene-level differences limit any claim that one growth rule is optimal across sensor and motion conditions; the three-seed comparisons are descriptive (full traces in `results/dilution_factorial/`).

Replica factorial and proportional budget (Tab. 3). We ran the same selector×throttle cells on two Replica scenes. Unlike TUM, coverage throttling lowers raw-error PSNR (-0.68 and -0.87 dB on `office0` and `office1`) and shrinks the map (113k to 65k and 84k to 40k); backlog throttling is worse still (-2.55 and -2.04 dB relative to coverage alone, with depth L1 up by 0.12 and 0.05 m). Without throttling, the temporal-influence selector trails raw error by 1.95 and 2.33 dB. Budget interventions help: a fixed ×4 budget adds $+2.81$ and $+1.56$ dB, and a budget proportional to map size (mean scale 3.67× and 3.87×, map size unchanged) adds $+2.88$ and $+1.56$ dB. On TUM the proportional budget adds $+3.73$ and $+2.99$ dB on `fr2_xyz` and `fr1_xyz` over the unthrottled baseline measured in the same batch (24.16 and 21.54 dB). These batches use the same code revision as Tabs. 1 and 2. **Dose-response (Tab. 4).** With free growth, PSNR rises monotonically with the budget multiplier on all four scenes (for example 24.24, 26.49, 27.82 dB on `fr2_xyz`), and the fraction of mature Gaussians updated per frame rises with it; their per-seed Spearman correlation over the three free-growth cells is $+1.00$ in every scene. Over all six cells it stays $+0.92$ and $+0.75$ on the TUM scenes, where a 23k cap is close to the natural map size, but not on Replica ($+0.20$ and -0.26), where the caps (60k, 40k) remove capacity and PSNR falls by 6–11 dB regardless of budget. Under a cap, ×2 and ×4 coincide once coverage reaches 1.00, because the budget exceeds what the

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.85	14.75	0.4655	67.2	21,128	+8.89 ✓
error_only	24.21	20.20	0.7023	11.0	72,366	+3.53 ✓
error_influence	25.38	22.35	0.7338	11.1	86,988	+2.36 ✓
error_influence_temporal	25.23	22.05	0.7315	10.9	86,187	+2.51 ✓
rtg_slam_reimpl	23.53	20.57	0.6833	10.1	60,050	+4.21 ✓
OURS	**27.74**	**28.73**	**0.7978**	**12.1**	**21,642**	—
FULL	*27.89*	*29.29*	*0.8036*	*11.3*	*70,295*	-0.15 ✓

Table 1. **Main benchmark, TUM `fr2_xyz`, 150 frames × 8 seeds.** All six controls differ significantly from OURS after Holm correction ($p_{\text{Holm}} = 0.0469$). FULL is 0.15 dB higher. Headroom recovery is 98.3% relative to no-op and 94.0% relative to the strongest budgeted baseline, at a 3.2× smaller map than FULL. Source: commit tag `paper-rerun-1`.

	TUM fr2_xyz			TUM fr1_xyz		
Selector / map control	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	24.13	.0205	73.1	21.52	.0378	79.1
Error, coverage	27.66	.0118	21.7	23.23	.0335	37.1
Error, coverage + backlog	27.67	.0117	21.5	23.32	.0334	35.1
Temporal×influence, no throttle	25.26	.0134	86.2	22.28	.0368	85.5
Temporal×influence, coverage	27.75	.0108	21.5	23.78	.0314	37.9
Temporal×influence, coverage + backlog	27.73	.0109	21.8	23.89	.0318	33.1
Error, optimization budget ×4	27.82	.0107	71.3	24.46	.0309	79.1
Error, hard map cap 23k	26.89	.0217	23.0	22.52	.1075	23.0
Error, hard map cap 39k	24.93	.0237	39.0	21.56	.1029	39.0

Table 2. **Corrected-pose factorial and dilution controls, 150 frames × 8 seeds per scene.** All factorial cells use the same modeled 15 ms optimization budget. The ×4 control uses 60 ms for optimization while keeping the original 15 ms spawn allowance. Depth L1 is in meters. Backlog is enabled independently of warmup only in the last throttle level. Per-run traces and paired intervals are in `results/rerun_paper1/dilution_factorial/`.

capped map can use. Update coverage therefore tracks quality wherever capacity is not binding. These are within-scene interventions on raw-error selection. They show that the system is compute limited and that the limit tightens as the map grows, but more compute improves any compute-limited optimizer, so they do not by themselves show that growth causes the limitation; the cap cells address that only on TUM.

Observed failure cases. Two TUM scenes show where the combined policy trails raw-error selection. *Exploration-heavy motion* (`fr1_room`, −1.04 dB) coincides with a smaller final map (92k vs 97k); the available run does not isolate whether throttling or selection causes the gap. *Rapid rotation* (`fr1_rpy`, −1.13 dB mean, finals 24.5 vs 24.1) is consistent with a lag from temporal smoothing, but this attribution has not been tested independently. **Horizon effect (Fig. 1):** at 30 frames OURS ties `error_only` (−0.01 dB on `fr2_xyz`); the gap appears at longer horizons. The new age-group logs in Tab. 2 link this divergence to a smaller update share for mature Gaussians in the growing map. Around frame 65 in seed 42, both error-only variants encounter difficult views; the unthrottled run continues spawning roughly 540–720 splats per frame and its PSNR remains low after frame 70, whereas the coverage run spawns roughly 60–130 (mean 92) and recovers by frame 75. The unthrottled loss rises from about 0.07 at frame 65 to 0.13 at frame 70 and stays high; opacity declines while the map expands. This is sustained divergence following shared hard views, not evidence for a single isolated gradient explosion. The logs cannot rule out floaters without a spatial geometry audit.

Scene dependence (Tab. 5). The combined policy wins the two confirmatory TUM comparisons ($n = 8$): `fr2_xyz` +3.53 dB and `fr1_xyz` +2.38 dB over raw-error selection, both $p = 0.0078$. It trails raw-error selection on all eight Replica scenes (−1.93 to −4.21 dB; `office0` and `office1` at $n = 8$, the rest exploratory at $n = 3$) and on `fr1_room` and `fr1_rpy`, so across 14 sequences it loses mean PSNR on 10. On Replica `office0`, raw-error selection with a weak throttle ($\kappa = 0.98$) reaches 30.17 dB versus 26.85 dB for the combined policy (supplement). The Replica factorial (Tab. 3) locates the loss: the unthrottled raw-error cell is the best of the six factorial cells on both scenes, adding the temporal-influence selector costs 1.95 and 2.33 dB, and coverage throttling costs 0.68 and 0.87 dB for raw-error selection. We cannot tell whether this reflects Replica's rendering, trajectories or depth quality. Compute allocation still matters there, since a map-proportional budget adds +2.88 and +1.56 dB. A simple hole-rate router

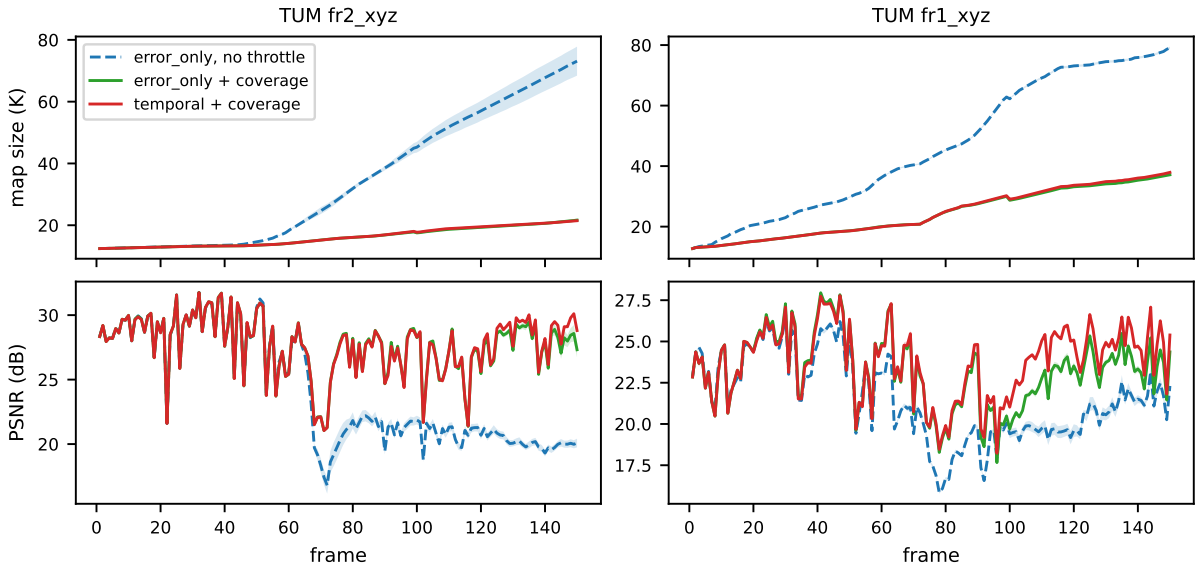


Figure 1. **Map growth and reconstruction quality over time** on TUM `fr2_xyz` and `fr1_xyz` (150 frames, eight seeds). Lines show the seed mean and shaded bands show ± 1 standard deviation. `error_only` versus `error + coverage` holds the selector fixed to show the coverage effect; `temporal + coverage` adds the combined policy. Paired sequence-level contrasts are reported in Tab. 2.

	Replica `office0`			Replica `office1`		
Condition	**PSNR**	**Depth L1**	**Map (k)**	**PSNR**	**Depth L1**	**Map (k)**
Error, no throttle	30.63	.0403	112.7	33.84	.0161	84.2
Error, coverage	29.95	.0556	65.2	32.97	.0273	39.9
Error, coverage + backlog	27.40	.1720	45.7	30.93	.0744	25.9
Temporal×influence, no throttle	28.68	.0489	115.8	31.51	.0202	85.8
Temporal×influence, coverage	28.14	.0651	66.3	31.73	.0329	42.0
Temporal×influence, coverage + backlog	26.18	.2385	45.8	29.79	.1093	24.1
Error, optimization budget ×4	33.44	.0326	107.5	35.41	.0106	82.8
Error, budget ∝ map size	33.51	.0327	107.5	35.41	.0108	82.7

Table 3. **Replica factorial and map-proportional budget, 150 frames × 8 seeds per scene.** Same cells as Tab. 2; all runs use the current code revision and a 15 ms modeled budget unless stated. The proportional cell sets the optimization budget to 15 ms × N_t/N_{ref} (mean scale 3.7–3.9×) with the spawn allowance fixed at 15 ms. Coverage and backlog throttling lower PSNR on Replica, whereas more optimization budget raises it.

that picks the better arm on the two datasets is in the supplement; its within-scene depth interventions did not reverse the ranking, so we do not use it as evidence for any mechanism.

Systems. Scheduler overhead is 2.54 ms; optimization-budgeted execution runs at about 10 FPS, so this is *not* end-to-end 15 ms real time. A separate baseline odometer (dense photometric+depth Gauss-Newton, no keyframes or loop closure) reaches ATE 0.9/2.0 cm on `fr2_xyz`/`fr1_xyz` over 100 frames (Supp. Tab. S3); it is a sanity check, not a SLAM claim. AlexNet LPIPS on `fr2_xyz` (150f×3) is 0.211 for OURS versus 0.294 for `error_only` (earlier code revision); depth L1 is in Tab. 2, and we report no mesh-based surface accuracy. Published-system numbers use different frames, resolution and views and give range context only (Supp. Tab. S2); short-horizon budget-curve and coverage-knee ablations are in Supp. Figs. S1–S2.

Reproducibility. Every table maps to a frozen bundle and a command in `paper_repro/`; the headline bundle records per-policy configs, source hashes, artifact hashes, pose protocol and clean-worktree status of each run set. Prior pose-bugged numbers (19.24 dB) are quarantined in `results/final_confirmation/DEPRECATED.md` and never compared. Environment: RTX 4050 Laptop GPU (6 GB), PyTorch 2.11+cu128, `gsplat` [20] 1.5.3, Python 3.12; peak allocation stays under 3.5 GB with the desktop compositor active, which is the intended deployment envelope for budgeted online mapping.

Scene (cap)	Free map growth ×1	Free map growth ×2	Free map growth ×4	Hard map cap ×1	Hard map cap ×2	Hard map cap ×4	ρ_{all}	ρ_{free}
`fr2_xyz` (23k)	24.24 (0.38)	26.49 (0.58)	27.82 (0.98)	26.90 (0.73)	27.85 (1.00)	27.85 (1.00)	+0.92	+1.00
`fr1_xyz` (23k)	21.53 (0.29)	22.78 (0.55)	24.45 (0.89)	22.52 (0.65)	23.63 (1.00)	23.63 (1.00)	+0.75	+1.00
`office0` (60k)	30.61 (0.17)	32.25 (0.34)	33.43 (0.72)	24.37 (0.24)	25.25 (0.49)	26.07 (1.00)	+0.20	+1.00
`office1` (40k)	33.85 (0.18)	34.88 (0.33)	35.42 (0.68)	22.79 (0.38)	22.85 (0.75)	22.90 (1.00)	-0.26	+1.00

Table 4. **Dose-response in optimization budget, 8 seeds.** Raw-error selection, no throttling; entries are mean PSNR (dB) with the mature-update coverage in parentheses, for a budget of $15 \times \{1, 2, 4\}$ ms with the spawn allowance fixed at 15 ms, with free growth or a hard Gaussian cap. ρ is the per-seed Spearman correlation between PSNR and mature-update coverage over the cells (all six, or the three free-growth cells), averaged over seeds. Once coverage reaches 1.00 extra budget is unused and PSNR saturates. On Replica the caps remove needed capacity (PSNR 6–11 dB lower at equal budget), so the cross-regime correlation does not hold there.

Scene	Frames	OURS	`error_only`	Δ	Map (k)	n
fr2_xyz (TUM)	150	27.74	24.21	+3.53 ✓	22/ 72	8
fr1_xyz (TUM)	150	23.47	21.09	+2.38 ✓	39 / 81	8
fr1_desk (TUM)	150	22.55	22.34	+0.21	27 / 56	3
fr1_room (TUM)	100	15.53	16.54	−1.01	92 / 97	3
fr1_rpy (TUM)	100	19.92	21.06	−1.14	45 / 62	3
fr3_static (TUM)	100	25.96	25.88	+0.08	13 / 18	3
office0 (Rep.)	150	26.85	30.17	−3.32 ✓	79 / 116	8
office1 (Rep.)	100	29.25	33.46	−4.21 ✓	42 / 74	8
office2 (Rep.)	99	22.08	24.67	−2.59	70 / 117	3
office3 (Rep.)	99	22.45	24.47	−2.02	77 / 150	3
office4 (Rep.)	99	23.58	25.93	−2.36	81 / 141	3
room0 (Rep.)	99	20.85	22.78	−1.93	82 / 150	3
room1 (Rep.)	100	20.42	24.29	−3.87	47 / 59	3
room2 (Rep.)	99	22.86	25.88	−3.03	78 / 146	3

Table 5. **Per-scene combined-policy breakdown, 14 sequences.** ✓ marks the reported significant comparisons at $n = 8$; other entries are exploratory. The combined policy wins the two confirmatory TUM comparisons but loses on both confirmatory Replica comparisons; remaining scenes have exploratory sample sizes.

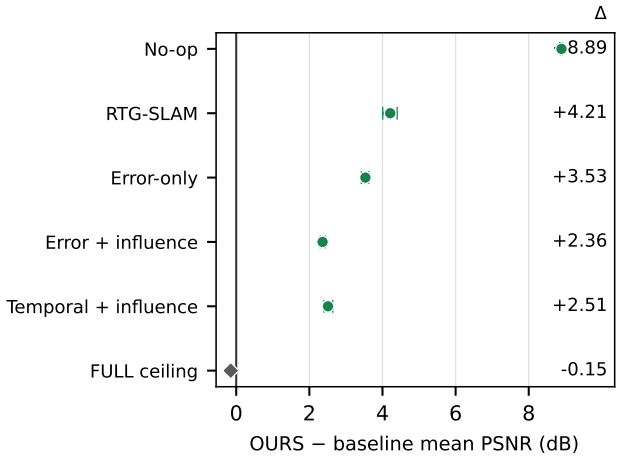


Figure 2. **Paired mean-PSNR differences** on fr2_xyz (150 frames, $n = 8$). Points show OURS − baseline; bars are 95% paired-bootstrap CIs. Positive favors OURS. FULL is 0.15 dB higher; all six contrasts pass Holm correction ($p = 0.0469$).

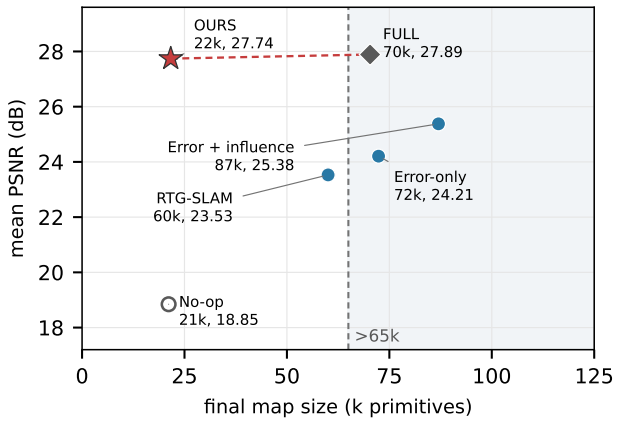


Figure 3. **Quality–map-size trade-off** on fr2_xyz (150 frames, $n = 8$). Among policies that optimize, only OURS and FULL lie on the frontier; FULL gains 0.15 dB with a 3.2× larger map. The four growing baselines hold 60–87k Gaussians; no-op stays compact at lower quality.

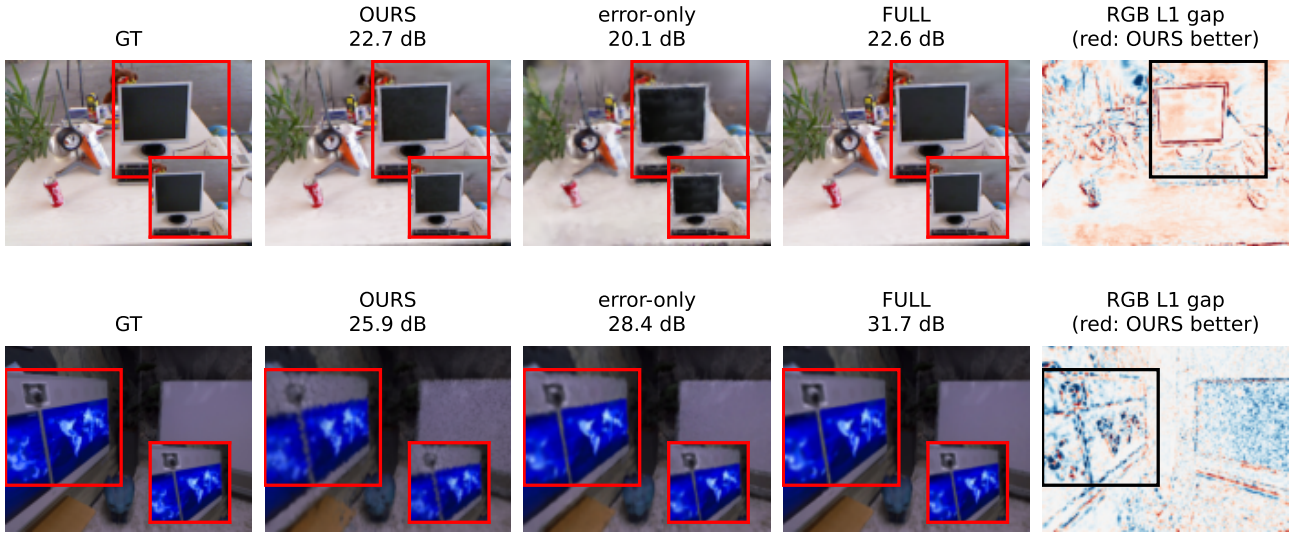


Figure 4. **Qualitative comparison across both evaluated substrates.** Top: an intermediate TUM `fr2_xyz` view (90), where OURS has lower per-view error than `error_only`; bottom: Replica `office0` view 98, where the ranking reverses. Columns show ground truth, OURS, `error_only`, FULL, and signed RGB L_1 error gap (red: OURS has lower error). Insets mark the same crop in each row. PSNR labels are per-view values from seed 42, shown only to illustrate the visual comparison; they are not sequence means.

5. Limitations

Scope of the mechanism. The budget interventions cover four scenes but one trajectory per scene and raw-error selection only. Coverage throttling helps on TUM and hurts on Replica, changes both spawn allowance and error thresholds, and map caps neither isolate map size nor remain valid on Replica, where they remove needed capacity. More budget improves any compute-limited optimizer, so the budget interventions show a compute limitation that tightens with map size, not that growth itself is harmful. Tabs. 1, 2, 3 and 4 share one code revision; Tab. 5 and the supplementary tables come from earlier revisions. Additional TUM scenes are exploratory ($n = 3$) and show smaller or depth-degrading effects. **Protocol.** Mapping uses ground-truth poses; horizons are at most 150 frames (about five seconds at 30 FPS); PSNR is train-view with only a same-trajectory holdout proxy. The 15 ms budget is modeled optimization time, not frame latency, and the pipeline runs at about 8–11 FPS on a laptop GPU. We do not compare against offline density-control methods [1, 7, 9, 22] or integrate the policy into a full SLAM system with its own tracker. **Statistics.** Eight seeds establish repeatability on fixed trajectories; the unit of generalization is the scene, and only two scenes are confirmatory. **Open items.** Longer sequences with a real tracker, comparisons against density-control baselines, additional scenes and sensors, and capacity-matched cap controls on Replica would test whether dilution generalizes beyond these four scenes.

6. Conclusion

Under a fixed modeled optimization budget, a growing map leaves established Gaussians under-updated. Raising the budget in proportion to map size improves raw-error selection on all four scenes tested ($+1.56$ to $+3.73$ dB), PSNR rises monotonically with budget, and the fraction of mature Gaussians updated per frame tracks quality wherever map capacity is not binding. The remedy of throttling spawns is scene dependent: it recovers several dB on the TUM scenes (27.74 dB on `fr2_xyz`, 0.15 dB below unconstrained optimization at a $3.2\times$ smaller map) but lowers PSNR on Replica, where the combined policy trails raw-error selection. We therefore present dilution as a compute-allocation mechanism with scene-dependent remedies, and leave open whether it persists on longer sequences and inside full SLAM systems.

References

[1] Guangchi Fang and Bing Wang. Mini-Splatting: Representing scenes with a constrained number of gaussians. In *ECCV*, 2024.

[2] Pengchong Hu and Zhizhong Han. VTGaussian-SLAM: RGBD SLAM for large scale scenes with splatting view-tied 3D gaussians. In *International Conference on Machine Learning (ICML)*, pages 24347–24370, 2025.

[3] Xinggang Hu, Chenyangguang Zhang, Mingyuan Zhao, Yuanze Gui, Xiangkui Zhang, and Xiangyang Ji. DyGS-SLAM: Real-time accurate localization and gaussian recon-

struction for dynamic scenes. In *ICCV*, pages 9561–9571, 2025.

[4] Huajian Huang, Longwei Li, Hui Cheng, and Sai-Kit Yeung. Photo-SLAM: Real-time simultaneous localization and photo-realistic mapping for monocular, stereo, and RGB-D cameras. In *CVPR*, 2024.

[5] Nikhil Keetha, Jay Karhade, Krishna Murthy Jatavallabhula, Gengshan Yang, Sebastian Scherer, Deva Ramanan, and Jonathon Luiten. SplaTAM: Splat, track & map 3D gaussians for dense RGB-D SLAM. In *CVPR*, pages 21357–21366, 2024.

[6] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023.

[7] Shakiba Kheradmand et al. 3D gaussian splatting as Markov chain Monte Carlo. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[8] Thai Luu, Quan Tran, Hieu Phan, and Tuan Dang. Geometry-aware online mapping for 3D gaussian splatting SLAM. *arXiv preprint arXiv:2608.14902*, 2026.

[9] Saswat Subhajyoti Mallick, Rahul Goel, Bernhard Kerbl, Markus Steinberger, Francisco Vicente Carrasco, and Fernando De La Torre. Taming 3DGS: High-quality radiance fields with limited resources. In *SIGGRAPH Asia Conference Papers*, 2024.

[10] Hidenobu Matsuki, Riku Murai, Paul H. J. Kelly, and Andrew J. Davison. Gaussian splatting SLAM. In *CVPR*, pages 18039–18048, 2024.

[11] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978.

[12] Zhexi Peng, Tianjia Shao, Yong Liu, Jingke Zhou, Yin Yang, Jingdong Wang, and Kun Zhou. RTG-SLAM: Real-time 3D reconstruction at scale using gaussian splatting. *ACM SIGGRAPH Conference Papers*, 2024.

[13] Erik Sandström, Yue Li, Luc Van Gool, and Martin R. Oswald. Point-SLAM: Dense neural point cloud-based SLAM. In *ICCV*, pages 18433–18444, 2023.

[14] Julian Straub, Thomas Whelan, et al. The Replica dataset: A digital replica of indoor spaces. *arXiv preprint arXiv:1906.05797*, 2019.

[15] Jürgen Sturm, Nikolas Engelhard, Felix Endres, Wolfram Burgard, and Daniel Cremers. A benchmark for the evaluation of RGB-D SLAM systems. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012.

[16] Maxim Sviridenko. A note on maximizing a submodular set function subject to a knapsack constraint. *Operations Research Letters*, 32(1):41–43, 2004.

[17] Peihao Wang et al. Steepest descent density control for compact 3D gaussian splatting. In *CVPR*, pages 26663–26672, 2025.

[18] Tianci Wen, Zhiang Liu, and Yongchun Fang. SEGS-SLAM: Structure-enhanced 3D gaussian splatting SLAM with appearance embedding. In *ICCV*, pages 28103–28113, 2025.

[19] Chi Yan, Delin Qu, Dan Xu, Bin Zhao, Zhigang Wang, Dong Wang, and Xuelong Li. GS-SLAM: Dense visual SLAM with 3D gaussian splatting. In *CVPR*, pages 19595–19604, 2024.

[20] Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for Gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025.

[21] Vladimir Yugay, Yue Li, Theo Gevers, and Martin R. Oswald. Gaussian-SLAM: Photo-realistic dense SLAM with gaussian splatting. *arXiv preprint arXiv:2312.10070*, 2023.

[22] Zheng Zhang, Wenbo Hu, Yixing Lao, Tong He, and Hengshuang Zhao. Pixel-GS: Density control with pixel-aware gradient for 3D gaussian splatting. In *ECCV*, 2024.

[23] Liyuan Zhu, Yue Li, Erik Sandström, Shengyu Huang, Konrad Schindler, and Iro Armeni. LoopSplat: Loop closure by registering 3D gaussian splats. In *International Conference on 3D Vision (3DV)*, 2025.